CHAPTER 55

Syphilis and Gonorrhoea

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Syphilis

Syphilis, a chronic systemic infection caused by Treponemapallidum subspecies pallidum, is usually sexually transmitted and is characterized by episodes of active disease interrupted by periods of latency. After an incubation period averaging 2-6 weeks, a primary lesion appears, often associated with regional lymphadenopathy. The secondary stage, associated with generalized mucocutaneous lesions and generalized lymphadenopathy, is followed by a latent period of subclinical infection lasting years or decades. Central nervous system (CNS) involvement may occur early in infection and may be symptomatic or asymptomatic. In about one-third of untreated cases, the tertiary stage appears, characterized by progressive destructive mucocutaneous, musculoskeletal, or parenchymal lesions; aortitis; or late CNS manifestations.

Etiology

The *Treponema* species include *T. pallidum* subspecies *pallidum*, which causes venereal syphilis.

Transmission and Epidemiology

Nearly all cases of syphilis are acquired by sexual contact with infectious lesions [i.e., the chancre, mucous patch, skin rash, or condylomatalata. Less common modes of transmission include nonsexual personal contact, infection in utero, blood transfusion, and organ transplantation.

Clinical Manifestations

Primary Syphilis : The typical primary chancre usually begins as a single painless papule that rapidly becomes eroded and usually becomes indurated, with a characteristic cartilaginous consistency on palpation of the edge and base of the ulcer. Multiple primary lesions are seen in a minority of patients. In heterosexual men the chancre is usually located on the penis, whereas in homosexual men it may be found in the anal canal or rectum, in the mouth, or on the external genitalia. In women, common primary sites are the cervix and labia. Consequently, primary syphilis goes unrecognized in women and homosexual men more often than in heterosexual men. Atypical primary lesions are common. The clinical appearance depends on the number of treponemes inoculated and on the immunologic status of the patient. A large inoculum produces a dark-field-positive ulcerative lesion in nonimmune volunteers but may produce a small dark-field-negative papule, an asymptomatic but seropositive latent infection, or no response at all in some individuals with a history of syphilis. A small inoculum may produce only a papular lesion, even in nonimmune individuals. Therefore, syphilis should be considered even in the evaluation of trivial or atypical dark-field-negative genital lesions. The genital lesions that most commonly must be differentiated from those of primary syphilis include those caused by herpes simplex virus infection, chancroid, traumatic injury, and donovanosis. Regional (usually inguinal) lymphadenopathy accompanies the primary syphilitic lesion, appearing within 1 week of lesion onset. The nodes are firm, nonsuppurative, and painless. Inguinal lymphadenopathy is bilateral and may occur with anal as well as with external genital chancres. The chancre generally heals within 4-6 weeks (range, 2-12 weeks), but lymphadenopathy may persist for months.

Secondary Syphilis

The protean manifestations of the secondary stage usually include mucocutaneous lesions and generalized nontender lymphadenopathy. The healing primary chancre may still be present in 15% of cases, and the stages may overlap more frequently in persons with concurrent HIV infection. The skin rash consists of macular, papular, papulosquamous, and occasionally pustularsyphilides; often more than one form is present simultaneously. The eruption may be very subtle, and 25% of patients with a discernible rash may be unaware that they have dermatologic manifestations. Initial lesions are pale red or pink, nonpruritic, discrete macules distributed on the trunk and proximal extremities; these macules progress to papular lesions that are distributed widely and that frequently involve the palms and soles. Rarely, severe necrotic lesions (luesmaligna) may appear; they are more commonly reported in HIV-infected individuals. Involvement of the hair follicles may result in patchy alopecia of the scalp hair, eyebrows, or beard in up to 5% of cases.

In warm, moist, intertriginous areas (commonly the perianal region, vulva, and scrotum), papules can enlarge to produce broad, moist, pink or gray-white, highly infectious lesions [condylomatalata (see Fig. e7-20)] in 10% of patients with secondary syphilis. Superficial mucosal erosions (mucous patches) occur in 10-15% of patients and commonly involve the oral or genital mucosa. The typical mucous patch is a painless silver-gray erosion surrounded by a red periphery.

Constitutional symptoms that may accompany or precede secondary syphilis include sore throat (15-30%), fever (5-8%), weight loss (2-20%), malaise (25%), anorexia (2-10%), headache (10%), and meningismus (5%). Acute meningitis occurs in only 1-2% of cases, but CSF cell and protein concentrations are increased in up to 40% of cases, and T. pallidum has been recovered from CSF during primary and secondary syphilis in 30% of cases; the latter finding is often but not always associated with other CSF abnormalities.

Less common complications of secondary syphilis include hepatitis, nephropathy, gastrointestinal involvement (hypertrophic gastritis, patchy proctitis, or a rectosigmoid mass), arthritis, and periostitis. Ocular findings that suggest secondary syphilis include pupillary abnormalities and optic neuritis as well as the classic iritis or uveitis. The diagnosis of secondary syphilis is often considered in affected patients only after they fail to respond to steroid therapy. Anterior uveitis has been reported in 5-10% of patients with secondary syphilis, and T. pallidum has been demonstrated in aqueous humor from such patients. Hepatic involvement is common in syphilis; although usually asymptomatic, up to 25% of patients may have abnormal liver function tests. Frank syphilitic hepatitis may be seen. Renal involvement usually results from immune complex deposition and produces proteinuria associated with an acute nephrotic syndrome. Like those of primary syphilis, the manifestations of the secondary stage resolve spontaneously, usually within 1-6 months.

Latent Syphilis

Positive serologic tests for syphilis, together with a normal CSF examination and the absence of clinical manifestations of syphilis, indicate a diagnosis of latent syphilis in an untreated person. The diagnosis is often suspected on the basis of a history of primary or secondary lesions, a history of exposure to syphilis, or the delivery of an infant with congenital syphilis. A previous negative serologic test or a history of lesions or exposure may help establish the duration of latent infection, which is an important factor in the selection of appropriate therapy. Early latent syphilis is limited to the first year after infection, whereas late latent syphilis is defined as that of 1 year's (or unknown) duration. T. pallidum may still seed the bloodstream intermittently during the latent stage, and pregnant women with latent syphilis may infect the fetus in utero. Moreover, syphilis has been transmitted through blood transfusion or organ donation from patients with latent syphilis. It was previously thought that untreated late latent syphilis had three possible outcomes: (1) persistent lifelong infection; (2) development of late syphilis; or (3) spontaneous cure, with reversion of serologic tests to negative. It is now apparent, however, that the more sensitive treponemal antibody tests rarely, if ever, become negative without treatment. Although progression to clinically evident late syphilis is very rare today, the occurrence of spontaneous cure is in doubt.

Involvement of the CNS

Traditionally, neurosyphilis has been

considered a late manifestation of syphilis, but this view is inaccurate. CNS syphilis represents a continuum encompassing early invasion (usually within the first weeks or months of infection), months to years of asymptomatic involvement, and, in some cases, development of early or late neurologic manifestations.

Other Manifestations of Late Syphilis

The slowly progressive inflammatory disease leading to tertiary disease begins early during infection, although these manifestations may not become clinically apparent for years or decades. Early syphilitic aortitis becomes evident soon after secondary lesions subside, and treponemes that trigger the development of gummas may have seeded the tissue years earlier.

Cardiovascular Syphilis

Cardiovascular manifestations, usually appearing 10-40 years after infection, are attributable to endarteritis obliterans of the vasa vasorum, which provide the blood supply to large vessels; T. pallidum DNA has been detected by PCR in aortic tissue. Cardiovascular involvement results in uncomplicated aortitis, aortic regurgitation, saccular aneurysm (usually of the ascending aorta), or coronary ostial stenosis. In the preantibiotic era, symptomatic cardiovascular complications developed in 10% of persons with late untreated syphilis, although syphilitic aortitis was demonstrated at autopsy in about one-half of African-American men with untreated syphilis. Today, this form of late syphilis is rarely seen in the developed world. Linear calcification of the ascending aorta on chest x-ray films suggests asymptomatic syphilitic aortitis, as arteriosclerosis seldom produces this sign. Syphilitic aneurysms-usually saccular, occasionally fusiform do not lead to dissection. Only 1 in 10 aortic aneurysms of syphilitic origin involves the abdominal aorta.

Late Benign Syphilis (Gumma)

Gummas are usually solitary lesions ranging from microscopic to several centimeters in diameter. Histologic examination shows a granulomatous inflammation, with a central area of necrosis due to endarteritis obliterans. Although rarely demonstrated microscopically, T. pallidum has been detected by PCR or recovered from these lesions, and penicillin treatment results in rapid resolution, confirming the treponemal stimulus for the inflammation. Common sites include the skin and skeletal system; however, any organ (including the brain) may be involved. Gummas of the skin produce indolent, painless, indurated nodular or ulcerative lesions that may resemble other chronic granulomatous conditions, including tuberculosis, sarcoidosis, leprosy, and deep fungal infections. Skeletal gummas most frequently involve the long bones, although any bone may be affected. Upper respiratory gummas can lead to perforation of the nasal septum or palate.

Congenital Syphilis

Transmission of T. pallidum across the placenta from a syphilitic woman to her fetus may occur at any stage of pregnancy, but fetal damage generally does not occur until after the fourth month of gestation, when fetal immunologic competence begins to develop. This timing suggests that the pathogenesis of congenital syphilis, like that of adult syphilis, depends on the host immune response rather than on a direct toxic effect of T. pallidum. The risk of fetal infection during untreated early maternal syphilis is 75-95%, decreasing to 35% for maternal syphilis of >2 years' duration. Adequate treatment of the woman before the 16th week of pregnancy should prevent fetal damage, and treatment before the third trimester should adequately treat the infected fetus. Untreated maternal infection may result in a rate of fetal loss of up to 40% (with stillbirth more common than abortion because of the late onset of fetal pathology), prematurity, neonatal death, or nonfatal congenital syphilis. Among infants born alive, only fulminant congenital syphilis is clinically apparent at birth, and these babies have a very poor prognosis. The most common clinical problem is the healthy-appearing baby born to a mother with a positive serologic test. Routine serologic testing in early pregnancy is considered cost-effective in virtually all populations, even in areas with a low prenatal prevalence of syphilis. Low-tech point-of-care tests are being developed to facilitate antenatal testing in resource-poor settings. Where the prevalence of syphilis is high or when the patient is at high risk of re-infection, serologic testing should be repeated in the third trimester and at delivery. Neonatal congenital syphilis must be differentiated from other generalized congenital infections, including rubella, cytomegalovirus or herpes simplex virus infection, and toxoplasmosis, as well as from erythroblastosisfetalis.

The manifestations of congenital syphilis include (1) early manifestations, which appear within the first 2 years of life (often at 2-10 weeks of age), are infectious, and resemble the manifestations of secondary syphilis in the adult; (2) late manifestations, which appear after 2 years and are noninfectious; and (3) residual stigmata. The earliest manifestations of congenital syphilis (appearing 2-6 weeks after birth) include rhinitis, or "snuffles" (23%); mucocutaneous lesions (35-41%); bone changes (61%), including osteochondritis, osteitis, and periostitis detectable by x-ray examination of long bones; hepatosplenomegaly (50%); lymphadenopathy (32%); anemia (34%); jaundice (30%); thrombocytopenia; and leukocytosis. CNS invasion by T. pallidum is detectable in 22% of infected neonates. Neonatal death is usually due to pulmonary hemorrhage, secondary bacterial infection, or severe hepatitis.

Late congenital syphilis (untreated after 2 years of age) is subclinical in 60% of cases; the clinical spectrum in the remainder of cases may include interstitial keratitis (which occurs at 5-25 years of age), eighth-nerve deafness, and recurrent arthropathy. Bilateral knee effusions are known as Clutton's joints. Asymptomatic neurosyphilis is present in about one-third of untreated patients, and clinical neurosyphilis occurs in one-quarter of untreated individuals >6 years old. Gummatousperiostitis occurs at 5-20 years of age and, as in nonvenereal endemic syphilis, tends to cause destructive lesions of the palate and nasal septum.

Classic stigmata include Hutchinson's teeth (centrally notched, widely spaced, peg-shaped upper central incisors), "mulberry" molars (sixth-year molars with multiple, poorly developed cusps), saddle nose, and saber shins.

Ancillary Investigations

Laboratory Examinations

Demonstration of the Organism : T. pallidum cannot be detected by culture. Historically, dark-field microscopy and immunofluorescence antibody staining have been used to identify this spirochete in samples from moist lesions such as chancres or condylomatalata, but these tests are rarely available today outside of research laboratories. More sensitive PCR tests have been developed but are not commercially available, although some laboratories perform in-house PCR testing.

T. pallidum can be found in tissue with appropriate silver stains, but these results should be interpreted with caution because artifacts resembling T. pallidum are often seen. Tissue treponemes can be demonstrated more reliably in research laboratories by PCR or by immunofluorescence or immunohistochemical methods using specific monoclonal or polyclonal antibodies to T. pallidum.

Serologic Tests for Syphilis

There are two types of serologic test for syphilis: nontreponemal and treponemal. Both are reactive in persons with any treponemal infection, including yaws, pinta, and endemic syphilis.

The most widely used nontreponemal antibody tests for syphilis are the rapid plasma reagin (RPR) and Venereal Disease Research Laboratory (VDRL) tests, which measure IgG and IgM directed against a cardiolipin-lecithin-cholesterol antigen complex. The RPR test is easier to perform and uses unheated serum; it is the test of choice for rapid serologic diagnosis in a clinical setting and can be automated. The VDRL test remains the standard for examining CSF. The RPR and VDRL tests are recommended for screening or for quantitation of serum antibody. The titer reflects disease activity, rising during the evolution of early syphilis and often exceeding 1:32 in secondary syphilis. After therapy for early syphilis, a persistent fall by fourfold or more (e.g., a decline from 1:32 to 1:8) is considered an adequate response. VDRL titers do not correspond directly to RPR titers, and sequential quantitative testing (as for response to therapy) must employ a single test.

Treponemal tests measure antibodies to native or recombinant T. pallidum antigens and include the fluorescent treponemal antibody-absorbed (FTA-ABS) test and the T. pallidum particle agglutination (TPPA) test, both of which are more sensitive for primary syphilis than the previously used hemagglutination tests. The T. pallidumhemagglutination (TPHA) test is widely used in Europe but is not available in the United States. When used to confirm positive nontreponemal test results, treponemal tests have a very high positive predictive value for diagnosis of syphilis. In a screening setting, however, these tests give false-positive results at rates as high as 1-2%. Treponemal tests are likely to remain reactive even after adequate treatment and cannot differentiate past from current T. pallidum infection.

Treponemalimmunochromatographic strip (ICS) tests and enzyme immunoassays (EIAs), based largely on reactivity to recombinant antigens, have also been developed. Treponemal EIAs have been approved as confirmatory tests and, because of their ease of automation, are now used for screening purposes by some large laboratories. Because treponemal tests cannot distinguish between current and treated syphilis or may be falsely reactive, clinicians may be uncertain about how to interpret reactive EIA screening results.

Considerable interest has recently been focused on point-of-care ICS tests that can be used in the field or in resource-poor settings. These treponemal tests are not yet approved for use in the United States but have been assessed in antenatal clinics in a number of developing countries.

Source - Harrisons Principles of Internal Medicine, 18th Edition and e medicine articles

Gonorrhea

Gonorrhea is a sexually transmitted infection (STI) of epithelium and commonly manifests as cervicitis, urethritis, proctitis, and conjunctivitis. If untreated, infections at these sites can lead to local complications such as endometritis, salpingitis, tuboovarian abscess, bartholinitis, peritonitis, and perihepatitis in female patients; periurethritis and epididymitis in male patients; and ophthalmia neonatorum in newborns. Disseminated gonococcemia is an uncommon event whose manifestations include skin lesions, tenosynovitis, arthritis, and (in rare cases) endocarditis or meningitis.

Microbiology

Neisseria gonorrhoeae is a gram-negative, nonmotile, non-spore-forming organism that grows singly and in pairs (i.e., as monococci and diplococci, respectively). Exclusively a human pathogen, the gonococcus contains, on average, three genome copies per coccal unit; this polyploidy permits a high level of antigenic variation and the survival of the organism in its host. Gonococci, like all other Neisseria species, are oxidase positive. They are distinguished from other neisseriae by their ability to grow on selective media and to utilize glucose but not maltose, sucrose, or lactose.

Clinical Features

Gonococcal Infections in Males : Acute urethritis is the most common clinical manifestation of gonorrhea in males. The usual incubation period after exposure is 2-7 days, although the interval can be longer and some men remain asymptomatic. Urethral discharge and dysuria, usually without urinary frequency or urgency, are the major symptoms. The discharge initially is scant and mucoid but becomes profuse and purulent within a day or two. Gram's stain of the urethral discharge may reveal PMNs and gram-negative intracellular monococci and diplococcic. The clinical manifestations of gonococcal urethritis are usually more severe and overt than those of nongonococcal urethritis, including urethritis caused by Chlamydia trachomatis; however, exceptions are common, and it is often impossible to differentiate the causes of urethritis on clinical grounds alone.

Gonococcal Infections in Females

Gonococcal Cervicitis : Mucopurulent cervicitis is the most common. Cervicitis may coexist with candidal or trichomonal vaginitis. N. gonorrhoeae primarily infects the columnar epithelium of the cervical os. Bartholin's glands occasionally become infected.

Women infected with N. gonorrhoeae usually develop symptoms. However, the women who either remain asymptomatic or have only minor symptoms may delay in seeking medical attention. These minor symptoms may include scant vaginal discharge issuing from the inflamed cervix (without vaginitis or vaginosis per se) and dysuria (often without urgency or frequency) that may be associated with gonococcal urethritis. Although the incubation period of gonorrhea is less well defined in women than in men, symptoms usually develop within 10 days of infection and are more acute and intense than those of chlamydial cervicitis.

Physical Examination

The physical examination may reveal a mucopurulent discharge (mucopus) issuing from the cervical os. Because Gram's stain is not sensitive for the diagnosis of gonorrhea in women, specimens should be submitted for culture or a nonculture assay. Edematous and friable cervical ectopy as well as endocervical bleeding induced by gentle swabbing are more often seen in chlamydial infection. Gonococcal infection may extend deep enough to produce dyspareunia and lower abdominal or back pain. In such cases, it is imperative to consider a diagnosis of pelvic inflammatory disease (PID) and to administer treatment for that disease.

Gonococcal Vaginitis

The vaginal mucosa of healthy women is lined by stratified squamous epithelium and is rarely infected by N. gonorrhoeae. However, gonococcal vaginitis can occur in anestrogenic women (e.g., prepubertal girls and postmenopausal women), in whom the vaginal stratified squamous epithelium is often thinned down to the basilar layer, which can be infected by N. gonorrhoeae. The intense inflammation of the vagina makes the physical (speculum and bimanual) examination extremely painful. The vaginal mucosa is red and edematous, and an abundant purulent discharge is present. Infection in the urethra and in Skene's and Bartholin's glands often accompanies gonococcal vaginitis. Inflamed cervical erosion or abscesses in nabothian cysts may also occur. Coexisting cervicitis may result in pus in the cervical os.

Anorectal Gonorrhea

Because the female anatomy permits the spread of cervical exudate to the rectum, N. gonorrhoeae is sometimes recovered from the rectum of women with uncomplicated gonococcal cervicitis. The rectum is the sole site of infection in only 5% of women with gonorrhea. Such women are usually asymptomatic but occasionally have acute proctitis manifested by anorectal pain or pruritus, tenesmus, purulent rectal discharge, and rectal bleeding. Among men who have sex with men (MSM), the frequency of gonococcal infection, including rectal infection, fell by 90% throughout the United States in the early 1980s, but a resurgence of gonorrhea among MSM has been documented in several cities since the 1990s. Gonococcal isolates from the rectum of MSM tend to be more resistant to antimicrobial agents than are gonococcal isolates from other sites. Gonococcal isolates with a mutation in mtrR (multiple transferable resistance repressor) or in the promoter region of the gene that encodes for this transcriptional repressor develop increased resistance to antimicrobial hydrophobic agents such as bile acids and fatty acids in feces and thus are found with increased frequency in MSM. This situation may have been responsible for higher rates of failure of treatment for rectal gonorrhea with older regimens consisting of penicillin or tetracyclines.

Pharyngeal Gonorrhea

Pharyngeal gonorrhea is usually mild or asymptomatic, although symptomatic pharyngitis does occasionally occur with cervical lymphadenitis. The mode of acquisition is oral-genital sexual exposure, with fellatio being a more efficient means of transmission than cunnilingus. Most cases resolve spontaneously, and transmission from the pharynx to sexual contacts is rare. Pharyngeal infection almost always coexists with genital infection. Swabs from the pharynx should be plated directly onto gonococcal selective media. Pharyngeal colonization with Neisseria meningitidis needs to be differentiated from that with other Neisseria species.

Ocular Gonorrhea in Adults

Ocular gonorrhea in an adult usually results from autoinoculation from an infected genital site. As in genital infection, the manifestations range from severe to occasionally mild or asymptomatic disease. The variability in clinical manifestations may be attributable to differences in the ability of the infecting strain to elicit an inflammatory response. Infection may result in a markedly swollen eyelid, severe hyperemia and chemosis, and a profuse purulent discharge. The massively inflamed conjunctiva may be draped over the cornea and limbus. Lytic enzymes from the infiltrating PMNs occasionally cause corneal ulceration and rarely cause perforation.

Prompt recognition and treatment of this condition are of paramount importance. Gram's stain and culture of the purulent discharge establish the diagnosis. Genital cultures should also be performed.

Gonorrhea in Pregnant Women, Neonates, and Children

Gonorrhea in pregnancy can have serious consequences for both the mother and the infant. Recognition of gonorrhea early in pregnancy also identifies a population at risk for other STIs, particularly chlamydial infection, syphilis, and trichomoniasis. The risks of salpingitis and PID-conditions associated with a high rate of fetal loss-are highest during the first trimester. Pharyngeal infection, most often asymptomatic, may be more common during pregnancy because of altered sexual practices. Prolonged rupture of the membranes, premature delivery, chorioamnionitis, funisitis (infection of the umbilical cord stump), and sepsis in the infant (with N. gonorrhoeae detected in the newborn's gastric aspirate during delivery) are common complications of maternal gonococcal infection at term. Other microorganisms and conditions, including Mycoplasma hominis, Ureaplasma urealyticum, C. trachomatis, and bacterial vaginosis (often accompanied by infection with Trichomonas vaginalis), have been associated with similar complications.

The most common form of gonorrhea in neonates is ophthalmia neonatorum, which results from exposure to infected cervical secretions during parturition. Ocular neonatal instillation of a prophylactic agent (e.g., 1% silver nitrate eyedrops or ophthalmic preparations containing erythromycin or tetracycline) prevents ophthalmia neonatorum but is not effective for its treatment, which requires systemic antibiotics.

Clinical features

The clinical manifestations are acute and usually begin 2-5 days after birth. An initial nonspecific conjunctivitis with a serosanguineous discharge is followed by tense edema of the eyelids, chemosis, and a profuse, thick, purulent discharge. Corneal ulcerations that result in nebulae or perforation may lead to anterior synechiae, anterior staphyloma, panophthalmitis, and blindness. Infections described at other mucosal sites in infants, including vaginitis, rhinitis, and anorectal infection, are likely to be asymptomatic. Pharyngeal colonization has been demonstrated in 35% of infants with gonococcal ophthalmia, and coughing is the most prominent symptom in these cases. Septic arthritis is the most common manifestation of systemic infection or DGI in the newborn. The onset usually comes at 3-21 days of age, and polyarticular involvement is common. Sepsis, meningitis, and pneumonia are seen in rare instances.

Gonococcal Arthritis (DGI)

DGI (gonococcal arthritis) results from

gonococcal bacteremia. In the 1970s, DGI occurred in "0.5-3% of persons with untreated gonococcal mucosal infection. The lower incidence of DGI at present is probably attributable to a decline in the prevalence of particular strains that are likely to disseminate. DGI strains resist the bactericidal action of human serum and generally do not incite inflammation at genital sites, probably because of limited generation of chemotactic factors. Strains recovered from DGI cases in the 1970s were often of the PorB.1A serotype, were highly susceptible to penicillin, and had special growth requirements-including arginine, hypoxanthine, and uracil-that made the organism more fastidious and more difficult to isolate.

Menstruation is a risk factor for dissemination, and approximately two-thirds of cases of DGI are in women. In about half of affected women, symptoms of DGI begin within 7 days of onset of menses. Complement deficiencies, especially of the components involved in the assembly of the membrane attack complex (C5 through C9), predispose to neisserial bacteremia, and persons with more than one episode of DGI should be screened with an assay for total hemolytic complement activity.

Clinical features

The clinical manifestations of DGI have sometimes been classified into two stages: a bacteremic stage, which is less common today, and a joint-localized stage with suppurative arthritis. A clear-cut progression usually is not evident. Patients in the bacteremic stage have higher temperatures, and chills more frequently accompany their fever. Painful joints are common and often occur together with tenosynovitis and skin lesions. Polyarthralgias usually include the knees, elbows, and more distal joints; the axial skeleton is generally spared. Skin lesions are seen in "75% of patients and include papules and pustules, often with a hemorrhagic component. The differential diagnosis of the bacteremic stage of DGI includes reactive arthritis, acute rheumatoid arthritis, sarcoidosis, erythema nodosum, drug-induced arthritis, and viral infections (e.g., hepatitis B and acute HIV infection). The distribution of joint symptoms in reactive arthritis differs from that in DGI, as do the skin and genital manifestations.

Laboratory Diagnosis

A rapid diagnosis of gonococcal infection in men may be obtained by Gram's staining of urethral exudates. The detection of gram-negative intracellular monococci and diplococci is usually highly specific and sensitive in diagnosing gonococcal urethritis in symptomatic males but is only "50% sensitive in diagnosing gonococcal cervicitis. Samples should be collected with Dacron or rayon swabs. Part of the sample should be inoculated onto a plate of modified Thayer-Martin or other gonococcal selective medium for culture. It is important to process all samples immediately because gonococci do not tolerate drying. If plates cannot be incubated immediately, they can be held safely for several hours at room temperature in candle extinction jars prior to incubation. If processing is to occur within 6 h, transport of specimens may be facilitated by the use of nonnutritive swab transport systems such as Stuart or Amies medium. For longer holding periods (e.g., when specimens for culture are to be mailed), culture media with self-contained CO2-generating systems (such as the JEMBEC or Gono-Pak systems) may be used. Specimens should also be obtained for the diagnosis of chlamydial infection.

PMNs are often seen in the endocervix on a Gram's stain, and an abnormally increased number (30 PMNs per field in five 1000x oil-immersion microscopic fields) establishes the presence of an inflammatory discharge. Unfortunately, the presence or absence of gram-negative intracellular monococci or diplococci in cervical smears does not accurately predict which patients have gonorrhea, and the diagnosis in this setting should be made by culture or another suitable nonculture diagnostic method. The sensitivity of a single endocervical culture is "80-90%. If a history of rectal sex is elicited, a rectal wall swab (uncontaminated with feces) should be cultured. A presumptive diagnosis of gonorrhea cannot be made on the basis of gram-negative diplococci in smears from the pharynx, where other Neisseria species are components of the normal flora.

Increasingly, nucleic acid probe tests are being substituted for culture for the direct detection of N. gonorrhoeae in urogenital specimens. A common assay employs a nonisotopic chemiluminescent DNA probe that hybridizes specifically with gonococcal 16S ribosomal RNA; this assay is as sensitive as conventional culture techniques. A disadvantage of non-culture-based assays is that N. gonorrhoeae cannot be grown from the transport systems. Thus a culture-confirmatory test and formal antimicrobial susceptibility testing, if needed, cannot be performed. Nucleic acid amplification tests (NAATs), including Roche Amplicor, Gen-Probe APTIMA Combo2 (which also detects Chlamydia), and BD ProbeTec ET, offer an advantage: urine samples can be tested with a sensitivity similar to that obtained when urethral or cervical swab samples are assessed by culture and other non-NAATs.

Because of the legal implications, the preferred method for the diagnosis of gonococcal infection in children is a standardized culture. Two positive NAATs, each targeting a different nucleic acid sequence, may be substituted for culture of the cervix or the urethra as legal evidence of infection; however, cervical specimens are not recommended for prepubertal girls. Nonculture tests for gonococcal infection have not been approved by the U.S. Food and Drug Administration for use with specimens obtained from the pharynx and rectum of infected children. Cultures should be obtained from the pharynx and anus of both girls and boys, the vagina of girls, and the urethra of boys. For boys with a urethral discharge, a meatal specimen of the discharge is adequate for culture. Presumptive colonies of N. gonorrhoeae should be identified definitively by at least two independent methods.

Blood should be cultured in suspected cases of DGI. The use of Isolator blood culture tubes may enhance the yield. The probability of positive blood cultures decreases after 48 h of illness. Synovial fluid should be inoculated into blood culture broth medium and plated onto chocolate agar rather than selective medium because this fluid is not likely to be contaminated with commensal bacteria. Gonococci are infrequently recovered from early joint effusions containing <20,000 leukocytes/L but may be recovered from effusions containing >80,000 leukocytes/L. The organisms are seldom recovered from blood and synovial fluid of the same patient.

Differentiate between Syphilis and Gonorrhea

S.No.	Syphilis	Gonorrhea
Etiology	The *Treponema* species include *T. pallidum* subspecies *pallidum*, which causes venereal Syphilis.	*N gonorrhoeae* is a gram-negative, intracellular, aerobic diplococcus; more specifically, it is a form of diplococcus known as the gonococcus
Transmission and Epidemiology	Nearly all cases of syphilis are acquired by sexual contact with infectious lesions [i.e., the chancre, mucous patch, skin rash, or condylomatalata. Less common modes of transmission include nonsexual personal contact, infection in utero, blood transfusion, and organ transplantation.	*N gonorrhoeae* is spread by sexual contact or through vertical transmission during childbirth.It mainly affects the host's columnar or cuboidal epithelium. Virtually any mucous membrane can be infected by this microorganism. Gonococcal infection usually follows mucosal inoculation during vaginal, anal, or oral sexual contact. It also may be caused by inoculation of mucosa by contaminated fingers or other objects. Transmission through penile-rectal contact is fairly efficient.

Sex	Men are affected more frequently with primary or secondary syphilis than women.	The physiologic ectopy of the squamocolumnar junction onto the ectocervix in the adolescent female is one factor that causes particular susceptibility to this infection.
Prognosis	The morbidity of syphilis ranges from the relatively minor symptoms of the primary stages of infection to the more significant constitutional systemic symptoms of secondary syphilis and the significant neurological and cardiovascular consequences of tertiary disease. Since latent syphilis can persist for years or decades, the manifestations of tertiary syphilis often occur much later in life, causing significant morbidity and even death if not identified and treated.	With adequate early therapy, complete cure and return to normal function are the rule. Most gonococcal infections respond quickly to cephalosporin therapy. Late, delayed, or inappropriate therapy may lead to significant morbidity or, on rare occasions, death.
Clinical Presentation	Primary syphilis occurs 10-90 days after contact with an infected individual.	The incubation period for gonorrhea is usually 2-7 days after exposure to an infected partner. In all patients who present with a possible STD, the history should include the following:Past history of STDs (including HIV infection and viral hepatitis)Treatment history for known STDsKnown symptoms of STDs in current or past sexual partnersType of contraception usedAny history of sexual assault
Clinical Presentation	It manifests mainly on the glans penis in males and on the vulva or cervix in females. Ten percent of syphilitic lesions are found on the anus, fingers, oropharynx, tongue, nipples, or other extragenital sites. Regional nontender lymphadenopathy follows invasion.	The most common site of gonococcal infection in women is the endocervix (80%-90%), followed by the urethra (80%), rectum (40%), and pharynx (10%-20%). If symptoms develop, they often manifest within 10 days of infection. Major symptoms include vaginal discharge, dysuria, intermenstrual bleeding, dyspareunia (painful intercourse), and mild lower abdominal pain. In men, urethritis is the major manifestation of gonococcal infection. Initial characteristics include burning upon urination and a serous discharge. A few days later, the discharge usually becomes more profuse, purulent, and, at times, blood-tinged.
Clinical Presentation	Secondary syphilisSecondary syphilis manifests in various ways. It usually presents with a cutaneous eruption within 2-10 weeks after the primary chancre and	

	is most florid 3-4 months after infection. The eruption may be subtle; 25% of patients may be unaware of skin changes.	
Investigations	Serologic testing is considered the standard method of detection for all stages of syphilis.	Culture is the gold standard test in the workup of gonorrheal infections. Bacterial culture has high sensitivity and high specificity, is relatively low cost, is suitable for different types of specimen sources, and can be used for epidemiologic and resistance testing purposes. Perform a culture or nonculture detection test for *N gonorrhoeae* on endocervical, urethral, pharyngeal, or rectal discharge. Because organisms are intracellular, attempt to obtain specimens in a manner that will contain mucosal cells and not merely discharge (similar to a Papanicolaou smear).
Investigations	In suspected acquired syphilis, the traditional approach has been to first perform nontreponemal serology screening using the Venereal Disease Research Laboratory (VDRL), rapid plasma reagin (RPR), or the recently developed ICE Syphilis recombinant antigen test	Based on CDC guidelines and the USPS Task Force recommendations, nucleic acid amplification test (NAAT) assay is the test of choice to evaluate for urogenital infections in both males and females, whether symptomatic or asymptomatic
Serologic tests	Sensitivity of the VDRL and RPR tests are estimated to be 78-86% for detecting primary syphilis, 100% for detecting secondary syphilis, and 95-98% for detecting tertiary syphilis.	These tests include latex agglutination, ELISA, immunoprecipitation, and complement fixation tests. Because of their lower sensitivity and specificity, especially in populations with a low prevalence of disease, these tests are not routinely used for diagnosis, but they can be used as adjuncts to the other laboratory tests and may help in making the diagnosis.

MCQs

1. What is the most common type of gonococcal infection seen in newborns?
   a. Liver infection
   b. Ocular infection (conjunctivitis)
   c. Mastitis
   d. Meningitis
2. Which of the following disease is an STD
   a. Gonorrhea b. Syphilis
   c. Leprosy d. Both (a) and (b)
3. Gonorrhea is caused by which microorganism
   a. Neisseria gonorrhoeae
   b. Ventral pallidium
   c. Treponemapallidum
   d. C. perfringes
4. Causative microorganism in Syphilis
   a. Ventral pallidium
   b. Treponemapallidum
   c. Neisseria gonorrhoeae
   d. Mycoplasma genitalium

5. Incubation period of Primary syphilis occurs after how many days of contact with an infected individual
   a. 10-90 days
   b. 2-7 days
   c. 7-14 days
   d. 20-60 days

6. Chancre formation seen in
   a. Gonorrhea
   b. Leprosy
   c. Syphilis
   d. AIDS

Answer

1.	b	2.	d	3.	a
4.	b	5.	a	6.	c